1 Work out an estimate for the value of $\frac{48.7 \times 61.2}{11.3}$

$$\frac{50 \times 60}{10} = \frac{3000}{10} = 300$$

300

(Total for Question 1 is 3 marks)

---

2 Work out an estimate for the value of $\frac{41.2 \times 19.8}{0.49}$

$$\frac{40 \times 20}{0.5} = \frac{800}{0.5} = 1600$$

1600

(Total for Question 2 is 3 marks)

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3 Work out an estimate for the value of $\frac{28.4 \times 21.05}{5.9}$

$$\frac{30 \times 20}{6} = \frac{600}{6} = 100$$

100

(Total for Question 3 is 3 marks)

4 Work out an estimate for the value of $\frac{7.4 + 23.05}{0.196}$

$$\frac{7+23}{0.2} = \frac{30}{0.2} = 150$$

$$\left[\text{OR } \frac{7+20}{0.2} = \frac{27}{0.2} = 135\right]$$

150

(Total for Question 4 is 3 marks)

5 Work out an estimate for the value of $\frac{91.25 \times 4.87}{2.31}$

$$\frac{90 \times 5}{2} = \frac{450}{2} = 225$$

225

(Total for Question 5 is 3 marks)

6 Work out an estimate for the value of $\frac{18.3 + 62.8}{0.13}$

$$\frac{20 + 60}{0.1} = \frac{80}{0.1} = 800$$

800

(Total for Question 6 is 3 marks)

7 Work out an estimate for the value of $\dfrac{21.75 + \sqrt{98.1}}{0.192}$

$$\frac{20 + \sqrt{100}}{0.2} = \frac{20+10}{0.2} = \frac{30}{0.2} = 150$$

150

(Total for Question 7 is 3 marks)

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8 Work out an estimate for the value of $\dfrac{8.3 \times 18.7}{0.52}$

$$\frac{8 \times 20}{0.5} = \frac{160}{0.5} = 320$$

320

(Total for Question 8 is 3 marks)

---

9 Eddie and Ellen use a calculator to work out $\dfrac{431.1}{14.3 + 3.8^2}$

Eddie's answer is 1.5
Ellen's answer is 15

One of those answers is correct.

Use approximations to find out which answer is correct.

$$\frac{400}{14+4^2} \quad \left[\text{or } \frac{400}{10+4^2}\right]$$

$$\frac{400}{14+16} = \frac{40\cancel{0}}{3\cancel{0}}$$

$$= 13.\dot{3}$$

Ellen's answer is correct.

(Total for Question 9 is 3 marks)

40

10 Ciara drives an average of 43.6 miles per week

52 weeks in a year

(a) Work out an estimate for the number of miles Ciara drives in a year.

$40 \times 50 = 2000$

2000 miles

(2)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

underestimate – I rounded down the miles per week and the number of weeks in a year

(1)

(Total for Question 10 is 3 marks)

8

11 Dennis gets paid £8.21 per hour he works.
Each week Dennis works 41 hours.

40

(a) Work out an estimate for the amount Dennis gets paid in a week.

$8 \times 40 = 320$

£ 320

(2)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

underestimate – I rounded down the pay and number of hours

(1)

(Total for Question 11 is 3 marks)

12 Mr Sykes wants to buy a calculator for every student in year 11.
There are 104 students in year 11. 100
Each calculator costs £6.05 6

(a) Work out an estimate for the amount of money Mr Sykes will spend on calculators.

100 × 6

£ 600

(2)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

underestimate – I rounded down the number of students and the cost.

(1)

(Total for Question 12 is 3 marks)

---

13 Phoebe's pays 2.8 pence per minute to use her phone. 3
On average Phoebe uses her phone for 77 minutes per day. 80

She pays he phone bill for 29 days. 30

(a) Work out an estimate for how much Phoebe pays.

3 × 80 × 30

80 × 90

7200 p

£ 72

(3)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

Overestimate – I rounded all the numbers up

(1)

(Total for Question 13 is 4 marks)

10

14 A circle has a radius of 11 metres.

(a) Work out an estimate for the area of the circle.

Area $= \pi r^2$

$= 3 \times 10^2$

$= 3 \times 100$

$= 300$

300 ..... m^2

OR 363 (3)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

underestimate – I rounded π and the radius down.

(1)

(Total for Question 14 is 4 marks)

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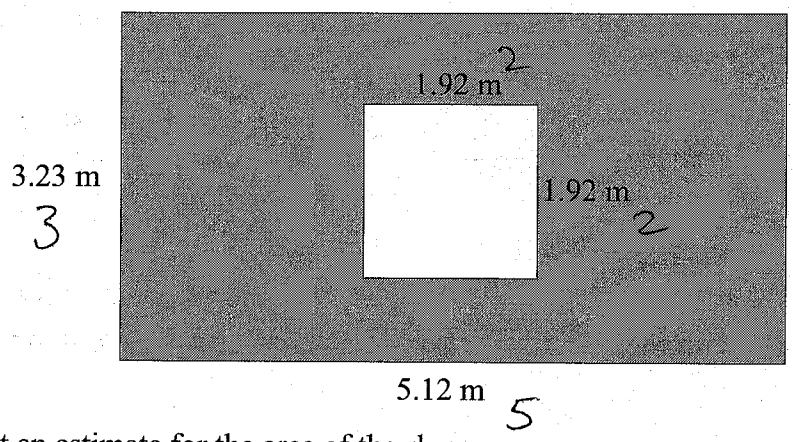
15 A shape is formed by cutting a square out of a rectangle.



(a) Work out an estimate for the area of the shape.

Big shape $3 \times 5 = 15\ m^2$

Small shape $2 \times 2 = 4\ m^2$

$15 - 4 = 11$

11 ..... m^2

(3)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

underestimate – the big shape will have a greater area and the small shape will have a smaller area.

(1)

(Total for Question 15 is 4 marks)

40

16 A baby was born every 43 seconds in the UK in 2018

~~(a)~~ Work out an estimate for the total number of babies born in the UK in 2018.
You must show how you get your answer.

$$\frac{\text{Number of seconds in a year}}{40}$$

Number of seconds in a year = $60 \times 60 \times 24 \times 365$

$\approx 60 \times 60 \times 20 \times 400$

3600×8000

$\approx 4000 \times 8000$

$32\,000\,000$

$$\frac{32\,000\,00\cancel{0}}{4\cancel{0}}$$

$800\,000$

800 000

(Total for Question 16 is 4 marks)

[600 000 to 900 000]